Nikhil Singh

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Skills

Languages: Rust, Haskell, Go, Python, Nix (Flakes), SQL, Java, Bash
Systems & Infra: NixOS, Kubernetes, Docker, Linux, AWS, GCP, Jenkins, GitHub Actions
Developer Tools: Git, Neovim/Nixvim, Tmux, QEMU, SSH
Datastores & Messaging: PostgreSQL, ClickHouse, Kafka, Redis/Valkey
AI Infrastructure: LLMs, MCP, LiteLLM, Embeddings, Vector Search, RAG
Engineering Focus: Distributed systems, reproducible builds, AI infrastructure, developer tooling

Experience

Product Engineer | *Juspay Technologies* Jun 2025 – Present

- Architected and engineered **Project Bernoulli**, a mission-critical Rust payment gateway bridging PSPs and client SDKs, processing 2M+ daily transactions at 99.99% uptime with end-to-end ownership of backend, infrastructure, CI/CD, and production support.
- Designed a multi-cloud deployment architecture spanning **AWS** and **GCP** with independent Kubernetes clusters and databases per cloud, achieving zero-downtime failover and zero committed-transaction loss with sub-3-minute replication lag.
- Engineered high-throughput event pipelines using **Kafka** and **ClickHouse** for real-time payment analytics, with sub-millisecond SDK session caching via **Redis/Valkey**.
- Built the **UDIR** (Unified Dispute and Issue Resolution) refund pipeline, automating dispute handling and accelerating merchant refund lifecycles.
- Built a layered **Jenkins** CI pipeline with cached compiler, dev-shell, and dependency layers published to an **Attic** binary cache (including GHC 8 compiler caching), reducing developer bootstrap from 20–40 min to 2–5 min and cutting CI build times by 40–60%; deployed a company-wide Nix devbox adopted by 800+ engineers.
- Designed the retrieval pipeline for **juspay-mcp**, vectorizing merchant documentation via **LiteLLM** embeddings to replace keyword search with semantic retrieval, reducing LLM context size and latency for natural-language API integration.
- Resolved a production deployment blocked by a stuck `ALTER TABLE` migration; diagnosed stale PostgreSQL planner statistics via `pg_stat_activity` and `EXPLAIN ANALYZE`, restoring service by running `ANALYZE` to switch the planner to an index scan.
- Authored a custom GPG library in Rust with MDC payload support for secure cryptographic operations in payment flows.

M.Tech Researcher – Systems Engineering | *Gautam Buddha University* May 2023 – May 2025

- Authored IEEE paper on modular NixOS configurations for reproducible OS management (IEEE Xplore).
- Co-authored IEEE paper on ML-based heavy rainfall prediction for the Northern Himalayan region.
- Developed Nix flake-based configurations using disko and home-manager for hardware-agnostic deployments.

Linux Systems Intern | *Decet Technology* 2023 – 2024

- Slashed deployment times from 2+ hours to under 5 minutes by engineering a custom Go-based CI/CD infrastructure, directly accelerating the backend release cycle.
- Built core Python backend services handling 10k+ daily requests with REST APIs and database integrations.

Package Maintainer – NixOS (nixpkgs) & Arch Linux | *NixOS & Arch Communities* 2020 – Present

- Upstreamed a highly adopted module to `home-manager`, enabling declarative user environment configurations for a large segment of the Nix community.
- Maintain multiple open-source Nix flake projects and 15+ packages across `nixpkgs` and the AUR with 50k+ combined downloads; host and manage a custom binary cache.
- Contributed 10+ upstream fixes to core ecosystem projects including NixOS and Hyprwm.

Projects

Ndots | *NixOS System Configuration (100+ Stars)* 2024 – Present

- Modular NixOS flake orchestrating reproducible configurations across multiple personal and organizational systems, supporting diverse machine architectures.
- Automated deployment pipelines integrating `sops` for secure secret management, reducing system setup time from days to under 30 minutes with full rollback capability.

Nvix | *Nixvim/Neovim Distribution for Terminal Users* 2024 – Present

- Maintained a high-performance Nixvim configuration with modular LSP/DAP/Treesitter plugin architecture supporting 20+ languages at sub-50ms startup, optimized for keyboard-centric workflows.

Tmux Minimal Status | *Terminal UI, Open Source* 2020 – Present

- Maintainer of popular tmux status bar (240+ GitHub stars); optimized for minimal resource footprint in high-frequency terminal environments.

Education

Gautam Buddha University Greater Noida, UP

Master of Technology in Software Engineering ***(CGPA: 8.83)*** *Aug 2023 – May 2025*

Bachelor of Technology in Computer Science Engineering *Aug 2020 – Aug 2023*